

Article

Advancing Sustainable Energy: Environmental and Economic Assessment of Plastic Waste Gasification for Syngas and Electricity Generation Using Life Cycle Modeling

Muhammad Hassan Javed ¹, Anees Ahmad ¹, Mohammad Rehan ², Farayi Musharavati ³,
Abdul-Sattar Nizami ^{1,4,*} and Mohammad Ilyas Khan ⁵

¹ Sustainable Development Study Centre, Government College University, Lahore 54000, Pakistan; hassanjavedduggal@gmail.com (M.H.J.); aneesjamal180@gmail.com (A.A.)

² Center of Excellence in Environmental Studies (CEES), King Abdulaziz University, Jeddah 21589, Saudi Arabia; dr.mohammad_rehan@yahoo.co.uk

³ Department of Mechanical and Industrial Engineering, Qatar University, Doha 2713, Qatar; farayi@qu.edu.qa

⁴ Graduate School of Energy and Environment, Korea University, Seoul 02481, Republic of Korea

⁵ Department of Chemical Engineering, College of Engineering, King Khalid University, Abha 61421, Saudi Arabia; mkaan@kku.edu.sa

* Correspondence: asnizami@gcu.edu.pk; Tel.: +92-3017729507

Abstract: The explosion of plastic waste generation, approaching 400 million tons per year, has created a worldwide environmental crisis that conventional waste management systems cannot handle. This problem can be solved through gasification, which converts nonrecyclable plastics to syngas with potential applications in electricity generation and synthetic fuel production. This study investigates whether syngas production from plastic waste by gasification is environmentally and economically feasible. Environmental impacts were assessed through a life cycle assessment framework using a life cycle impact assessment approach, ReCiPe 2016, with 10 midpoint/endpoint categories. Midpoint results of the baseline scenario with grid-mix electricity revealed climate change (GWP) of 775 kg CO₂ equivalent and fossil depletion potential (FDP) of 311 kg oil equivalent per ton of plastic waste. Meanwhile, a solar scenario showed GWP as 435 kg CO₂ equivalent and FDP as 166 kg oil equivalent per ton of plastic waste. Switching to solar energy cut GWP 44% and FDP 47%, respectively. However, the tradeoffs were higher human toxicity potential (HTP) and marine ecotoxicity potential (METP) due to upstream material extraction of renewable systems, respectively. Among environmental performance drivers, electricity inputs and operating materials were identified through sensitivity and uncertainty analyses. Syngas production from a plant of 50 tons per day can yield electricity sales revenue of USD 4.79 million, excluding USD 4.05 million in operational expenditures. Financial indicators like a 2.06-year payback period, USD 5.32 million net present value over a 20-year project life, and 38.2% internal rate of return indicate the profitability of the system. An external cost analysis showed emissions-related costs of USD 26.43 per ton of plastic waste processed, dominated by CO₂ and NO_x emissions. Despite these costs, the avoided impacts of less landfilling/incineration and electricity generation support gasification. Gasification should be promoted as a subsidy and incentive by policymakers for wider adoption and integration into municipal waste management systems. Findings show it can be adapted to global sustainability goals and circular economy principles while delivering strong economic returns. The study findings also contribute to several Sustainable Development Goals (SDGs), for instance, SDG 7 by promoting clean energy technologies, SDG 12 by implementing circular economy, and SDG 13 by reducing greenhouse gas (GHG) emissions.



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1. Introduction

Plastic is a major inorganic solid waste found in our daily municipal solid waste (MSW) production. Rising population, rapid urbanization, and industrial growth are key factors creating serious artificial problems that ultimately threaten the existence of life on Earth [1]. Plastic production surged from 2 million tons in 1950 to over 400 million tons annually by 2022, reflecting an exponential 180-fold increase in consumption between 1950 and 2018. This rapid growth has created an unprecedented waste management crisis [2]. Plastics, famous for their versatility and durability, have simultaneously become persistent pollutants due to their nonbiodegradability, resulting in major environmental and societal challenges, including pollution, greenhouse gas (GHG) emissions, and threats to biodiversity [3]. Conventional waste management strategies like landfilling and incineration worsen these issues rather than reducing them [4].

Landfilling, which accounts for over 60% of worldwide plastic waste, causes soil and groundwater contamination through toxic leachates and methane emissions [5]. Similarly, incineration reduces waste volume but releases harmful emissions like dioxins and furans that threaten human health and the environment [6]. Additionally, traditional recycling approaches are constrained by plastic waste heterogeneity and contamination, leaving a significant portion of plastic waste unmanageable. Waste-to-energy (WTE) technologies, such as gasification, provide practical solutions to both waste management and energy generation challenges [7,8].

Plastic waste is one of the most persistent and difficult waste streams to manage globally, exacerbated by the excessive quantities of nonrecyclable plastics, including mixed and contaminated waste [9]. These plastics, often composed of multiple polymers or infused with additives, are challenging to process using conventional recycling techniques. Mechanical recycling is typically limited to clean, single-polymer waste. At the same time, contamination with organic matter or incompatible substances further complicates recycling, leading to reliance on less sustainable methods such as incineration or landfilling [10–12].

Plastics' environmental consequences are severe and multifaceted. Being non-biodegradable, plastics persist in the environment for centuries, with a significant portion entering marine ecosystems as microplastics. These microplastics infiltrate the food chain, threatening biodiversity and human health [13,14]. Additionally, the degradation of plastics under sunlight or chemical stress releases hazardous substances, further exacerbating environmental risks [15].

Plastic waste management also contributes significantly to GHG emissions. Plastics derived from fossil fuels have high carbon footprints across their life cycle. Landfills emit methane, a potent GHG, as organic waste decomposes, while incineration releases large quantities of CO₂ despite reducing waste volume [16,17]. These processes intensify climate change and hinder progress toward sustainable waste management systems.

A lack of infrastructure for advanced recycling and WTE technologies remains a major barrier to effective plastic waste management, particularly in developing regions. Many such regions rely on rudimentary collection and disposal systems, with plastics often ending up in open dumps or being openly burned, releasing toxic pollutants [18–20]. Even in regions with advanced systems, facilities capable of processing mixed and contaminated plastics still need to be improved, creating bottlenecks for sustainable waste solutions.

Despite these challenges, innovative approaches to plastic waste management, such as the valorization of nonrecyclable plastics, offer significant opportunities [21]. Gasification stands out as a promising solution for converting plastic waste into syngas, an intermediate product with diverse applications. Operating under controlled oxygen-limited conditions, gasification minimizes the production of toxic byproducts like dioxins and furans, making it a cleaner alternative to incineration [22].

Plastic waste can be valorized through various approaches, including mechanical recycling, thermochemical conversion like pyrolysis, gasification, and biological approaches. Mechanical recycling has limitations of feedstock contamination, whereas thermochemical methods like pyrolysis and gasification can manage mixed and contaminated plastic waste [23]. Gasification operates under oxygen-limited conditions that convert plastic waste into syngas. Biological approaches are environment-friendly but constrained by slow reaction rates and limited to specific plastic types. Despite these advancements, there remains a significant gap in evaluating the environmental and economic feasibility of gasification on a system-wide scale [24]. This study addresses this gap by integrating LCA and economic analysis to evaluate syngas production from plastic waste under various energy scenarios.

Gasification offers an effective pathway for managing nonrecyclable plastics, transforming waste into a valuable resource while reducing landfill dependency. Syngas, primarily composed of hydrogen (H_2) and carbon monoxide (CO), can serve as a precursor for synthetic fuels, electricity generation, and chemical production [25]. This aligns with the principles of the circular economy by closing resource loops and promoting sustainable resource use. Furthermore, the integration of gasification systems into existing waste management frameworks can significantly reduce environmental pollution and enhance energy security [26].

Advanced gasification systems, such as integrated gasification combined cycle (IGCC) technologies, combine syngas production with electricity generation in highly efficient turbines, achieving optimal energy recovery. Emerging technologies, such as plasma and supercritical water gasification, further enhance the potential of this process [27]. Plasma gasification uses ionized gases at high temperatures to break down plastics at the molecular level, while supercritical water gasification processes wet plastic waste without the need for energy-intensive drying. Both technologies reduce emissions and maximize syngas purity, demonstrating adaptability to various waste types [28].

Gasification also creates economic opportunities. Syngas production generates revenue streams through the sale of synthetic fuels and electricity, while byproducts like char can be marketed as activated carbon or soil enhancers. Combined with renewable energy inputs, advanced gasification systems can achieve cost reductions and improved environmental performance, making them competitive with traditional energy sources [29]. With strong policy support, including subsidies, tax credits, and collaboration between researchers and industry, gasification can become a cornerstone of sustainable plastic waste management [30].

Gasification is a thermochemical process that transforms nonrecyclable plastics into synthesis gas (syngas), a valuable product composed primarily of hydrogen (H_2), carbon monoxide (CO), and methane (CH_4). The process operates under oxygen-limited conditions, which reduces harmful byproducts and maintains precise control over syngas composition [31]. Gasification can process mixed and contaminated plastic waste streams, providing an alternative to conventional disposal methods. Furthermore, syngas is a versatile energy carrier that can be utilized for electricity generation, synthetic fuel production, and chemical manufacturing [32]. This makes gasification a key technology for transition-

ing towards sustainable energy systems while reducing dependency on virgin fossil fuels and promoting circular economy principles [33].

Beyond its waste management potential, gasification aligns with the United Nations Sustainable Development Goals (SDGs), particularly SDG 7 (Affordable and Clean Energy), SDG 12 (Responsible Production and Consumption), and SDG 13 (Climate Action). The process reduces landfill dependency and generates clean energy, thus contributing to plastic pollution mitigation and global energy security [34]. Syngas production from plastic waste also has economic benefits, including avoiding the costs of landfilling and incineration and the potential for profitable byproducts. Advanced systems with optimized operational parameters can further enhance the economic feasibility of gasification, making it a competitive alternative in waste management and energy production [35].

This study aims to evaluate the feasibility of plastic waste gasification as a viable solution for syngas and electricity generation from both economic and environmental perspectives. A life cycle assessment (LCA) framework is employed to evaluate process environmental impacts across various life cycle impact categories. Additionally, an economic evaluation of syngas utilization in energy markets, considering internal costs (IC), e.g., operational expenses, and external costs (EC), e.g., environmental damage, is conducted. The research follows a cradle-to-gate approach, encompassing plastic waste collection, preprocessing, gasification, and syngas utilization.

2. Materials and Methods

The LCA approach was used to investigate the environmental impacts and techno-economic assessment of syngas production from plastic waste. According to the International Organization for Standardization (ISO) guidelines and specific requirements 14040–14044 standard series [36,37], there are four main parts in any LCA study: (i) the goal and scope, (ii) the life cycle inventory analysis, (iii) the life cycle impact assessment, and (iv) the detailed interpretation described below.

2.1. Goal and Scope Definition

The goal of this study is to evaluate the environmental impacts and economic feasibility of plastic waste gasification for syngas production. Using an LCA framework, this study quantified the environmental performance of the gasification process across different impact categories such as climate change, land use, and freshwater ecotoxicity. This study also included an economic analysis of the financial viability of the process, focusing on operational and ECs relating to environmental impacts. This dual evaluation aims to provide an overview of plastic waste gasification as a possible solution for WTE conversion.

This study is designed on a cradle-to-gate LCA framework, encompassing all stages from plastic waste collection and preprocessing to syngas utilization. Plastic waste processing includes sorting/shredding/drying to prepare the feedstock for gasification. Gasification itself is a controlled thermochemical reaction that changes plastic polymers into syngas. One metric ton of plastic waste processing was selected as the study's functional unit. This benchmark enables a standardized comparison of environmental and economic impacts under different technological and operational scenarios. The analysis is simplified by assuming that syngas composition remains constant in all modeled scenarios. In addition, the analysis assumes that the gasification system optimally operates according to the capabilities of commercial-scale technologies today.

2.2. Gasification Process

Gasification is a thermochemical process that converts carbonaceous substances, such as plastic waste, into syngas through partial oxidation at high temperatures (800–1200 °C).

Figure 1 shows the typical gasification technology. This process involves pyrolysis, where heat breaks down polymers into char, tar, and volatile gases [25]. The volatile gases undergo partial oxidation and reduction reactions, producing hydrogen (H_2), carbon monoxide (CO), methane (CH_4), and carbon dioxide (CO_2), along with minor impurities [12]. Key reactions (Equations (1)–(5)) include:

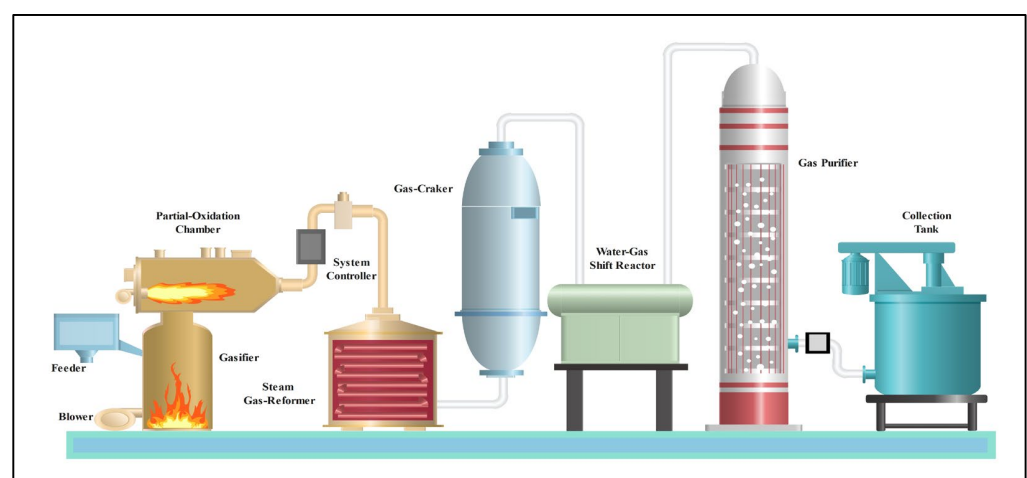
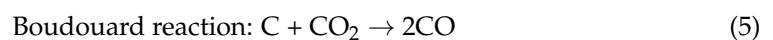
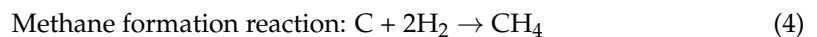
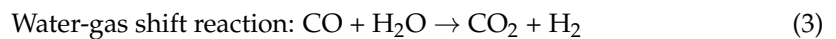
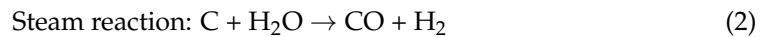
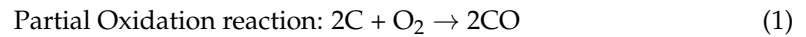


Figure 1. Technological diagram of plastic waste gasification.

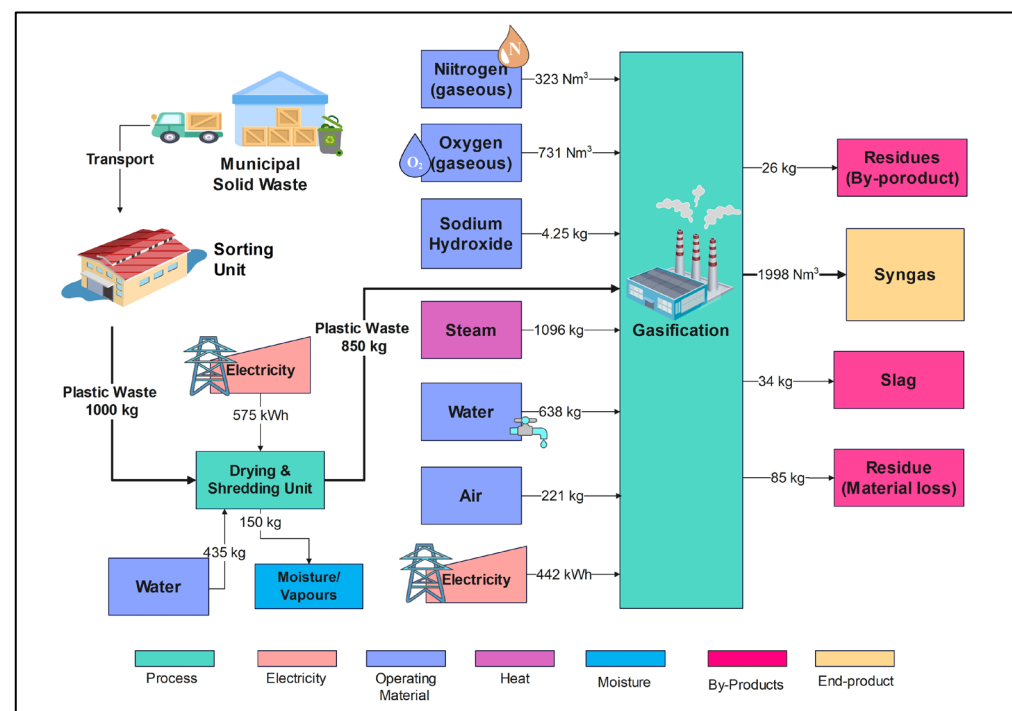
2.3. Life Cycle Inventory (LCI) Analysis

Input and output gasification data from primary and secondary sources were collected in LCI phase. Primary data are derived from experimental and simulation studies of syngas production, key data being reported by Chen et al. [38], which used original input/output data of gasification from Japan Containers & Packaging Recycling Association [39]. Some primary and secondary data are imported from Ecoinvent and GaBi databases, respectively, to strengthen the robustness of the inventory. Together, these sources provide a dataset covering material and energy flows in plastic waste gasification, as given in Table 1. Data related to these input materials and output products are included in the inventory table.

The inventory includes all major inputs for the gasification process, like plastic feed-stock ratio, energy consumption as electricity, and operating materials, including air, oxygen, nitrogen, water, sodium hydroxide, and steam, that are necessary for the basic gasification process. Outputs include the main product, syngas, byproducts like slag and char, and emissions (CO_2 , NO_x , SO_2 , etc.) from air, water, and soil. Input and output flows of the gasification process are shown in Figure 2.

Table 1. Life cycle inventory of syngas production inputs and outputs [38].

Shredding and Drying				
Input/Output		Item	Amount	Units
Input	Material	Waste Plastic	1000	kg
	Utility	Electricity	575	kWh
		Water	435	kg
Output	Material	Plastic granulates	850	kg
	Loss	Waste (unspecified)	150	kg
Gasification				
Input/Output		Item	Amount	Units
Input	Material	Waste Plastic	850	kg
	Utility	Oxygen	731	Nm ³
		Caustic soda	4.25	kg
		Electricity	442	kWh
		Steam	1096.5	kg
		Air	221	Nm ³
		Nitrogen	323	Nm ³
		Water	637.5	kg
Output	Product	Syngas	1998	Nm ³
	Byproduct	Slag	34	kg
		Residue	26	kg
	Residue	Material loss	85	kg

**Figure 2.** Energy and mass balance of the gasification process.

2.4. Life Cycle Impact Assessment (LCIA)

The LCIA in this study was examined at the midpoint and endpoint by GaBi software using the ReCiPe 2016 (H) approach (Version 1.1) [40]. The ReCiPe2016 method is an updated version of ReCiPe2008, which is frequently used in LCA studies [41,42]. The Recipe 2016 approach was selected due to harmonized characterization factors at midpoints and endpoints, while other LCIA approaches evaluate either the midpoint or endpoint [43].

The hierarchical (H) perspective of ReCiPe 2016 was applied with a balanced weighting of current scientific consensus on impact pathways.

2.4.1. Selected Impact Categories

The ReCiPe 2016 (H) approach contains 18 midpoint categories from which environmental impacts can be analyzed. This approach is chosen because it links midpoint impacts, such as emissions and resource use, with longer-term impacts on human health, ecosystems, and resource availability. The hierarchical (H) perspective is applied due to the balanced weighting of the current scientific consensus on impact pathways. In this study, for the midpoint assessment, 10 categories out of 18 are used in the ReCiPe 2016 methodologies. Selected categories are climate change as global warming potential (GWP), fine particulate matter formation (FPMFP), fossil depletion (FDP), freshwater consumption (FWCP), human toxicity (HTP), land use (LUP), marine ecotoxicity (METP), photochemical ozone formation (POFP), terrestrial acidification (TAP), and terrestrial ecotoxicity (TETP). These categories were selected for their relevance to plastic waste gasification processes and environmental sustainability in general.

Endpoint assessment was employed to supplement midpoint categories by aggregating impacts into three broad categories: ecosystem, human health, and resource depletion. These endpoint indicators give an overview of the long-term consequences of environmental interventions to enable stakeholder evaluation of tradeoffs and prioritization of mitigation efforts [44]. It combines midpoint indicators with endpoint indicators to provide an overview of the environmental impacts of gasification. Normalized results are also evaluated to compare and find out which categories resulted in high emissions and impacts.

2.4.2. Scenario Design

Two main scenarios have been created to compare the environmental performance of plastic waste gasification under different conditions: the baseline (grid mix electricity) scenario and the solar (renewable electricity) scenario. The baseline scenario adopted grid mix electricity for all processes involved in gasification. Pakistan's grid mix electricity consists of 25.4% hydel, 8.4% nuclear, 6.8% renewable, and 59.4% thermal [45]. It serves as a benchmark to evaluate conventional gasification system emissions as well as energy related to processing plastic wastes. Solar energy was employed in the solar scenario to evaluate the difference in environmental impact or fluctuation due to renewable energy replacement. The renewable energy scenario replaced grid mix electricity with solar electricity for gasification processes to explore its potential to lower the carbon footprint and enhance the sustainability of the gasification system. Aligning with global efforts transitioning towards more renewable sources, this renewable energy scenario gives insight into environmental advantages gained through integration between renewable energies and waste management technologies. Both scenarios were modeled using identical system boundaries and functional units to facilitate comparative evaluation. This provided useful insights for optimizing gasification processes while increasing their sustainability.

2.5. Interpretation

The interpretation phase of LCA involved different analyses, assumptions, and results in pinpointing critical environmental hotspots, evaluating tradeoffs, and proposing recommendations to optimize sustainability [43]. Contribution analyses were performed to check the main contributors' relative impacts. There were two categories for contribution analysis: electricity and operating materials, which include oxygen, nitrogen, water, sodium hydroxide, steam, and air. This analysis gave valuable insights into optimizing processes to meet both environmental and economic outcomes simultaneously. Finally,

recommendations are offered to combat identified impacts and enhance sustainability in gasification systems.

Uncertainty Analysis

To test the robustness of the results, uncertainty analysis was performed using sensitivity analysis and Monte Carlo analysis. Sensitivity analysis was performed using varying critical gasification process parameters like electricity, oxygen, nitrogen, water, air, sodium hydroxide, and steam. These parameters were adjusted by a 10% increase and a 10% decrease to identify their impact on the environmental and economic performance of the system. Increasing/decrementing these contributors allows for the analysis of the effect of operational variability on performance indicators and selected impact categories. Monte Carlo analysis was performed by decreasing 10% and increasing 10% major parameters like electricity, oxygen, nitrogen, water, air, sodium hydroxide, and steam.

2.6. Life Cycle Costing

A life cycle cost (LCC) assessment was conducted using LCA, having the same system boundary and functional unit. The LCC focused on evaluating the economic feasibility of plastic waste gasification, incorporating both IC and EC. The LCC framework accounts for the conventional costs associated with the gasification process and the environmental costs of emissions generated during the life cycle stages. This dual approach enables a comprehensive understanding of the process's financial and environmental implications. This LCC approach was adopted by [46,47].

The LCC was calculated using the equation:

$$\text{LCC} = \text{IC} + \text{EC} \quad (6)$$

where IC illustrates the internal costs and EC refers to the external costs. This approach facilitates the identification of opportunities to reduce the total LCC, thereby achieving a balance between economic benefits for producers and public interest.

2.6.1. Internal Costs

IC involves two main components: initial capital investment (ICI) costs and operational costs. ICI includes the costs associated with equipment procurement and construction, while operational cost encompasses various recurring costs such as raw materials, utilities, labor, maintenance, management, transportation, and taxes. The IC was calculated using the following equation:

$$\text{IC} = \text{Cd} + \text{Cr} + \text{Cu} + \text{CM} + \text{Cl} + \text{Cm} + \text{Ctr} \quad (7)$$

In this equation, Cd is the depreciation cost, which was calculated using the following method as shown below:

$$\text{Cd} = \text{ICI} (1 - 5\%) / 20 \quad (8)$$

Here, 5% represents the salvage value ratio and 20 years is the assumed lifetime of the gasification project. Cr is the cost of raw materials and Cu represents the utility costs, respectively. These were determined using data derived from the literature and relevant industrial documents. Management costs CM were estimated based on operational data specific to plastic waste gasification facilities. Labor costs Cl were calculated using average salary data obtained from relevant industry sources, while maintenance costs Cm were estimated using figures from the previously published literature. Taxes Ct were included as part of the operational expenditure.

2.6.2. External Costs

The EC represents the potential environmental damage costs associated with emissions from the gasification process. EC is calculated using a simplified equation that quantifies the marginal damage cost based on the willingness to avoid environmental harm [46]. The equation used for EC computation is:

$$EC = \sum_{k=1}^7 C_k \times E_{k,lc} \quad (9)$$

In this equation, C_k represents the cost coefficient of emission k , which includes CO_2 , CH_4 , NO_x , CO , $NMVOC$, PM , and SO_2 . The coefficients were derived from the published literature [47]. $E_{k,lc}$ represents the life cycle amount of emission k , determined using inventory data modeled in GaBi software (Version 10.0.0.71). This software facilitated the calculation of emissions for each life cycle stage, with detailed values provided in associated emission tables. By integrating IC and EC into the LCC framework, this study provides a comprehensive evaluation of the economic implications of plastic waste gasification.

3. Results

3.1. Midpoint Assessment

Midpoint results were evaluated after analyzing the potential of 10 selected categories: GWP, FPMFP, FDP, FWCP, HTP, LUP, METP, POFP, TAP, and TETP, as shown in Table 2. The table highlights the environmental impacts of syngas production from plastic waste through the gasification process, emphasizing several categories. It discusses emissions contributing to GWP (7.75×10^2 kg CO_2 eq.) and PMFP (2.84×10^{-1} kg $PM_{2.5}$ eq.), noting the potential for improvement through reduced fossil energy inputs and emission control systems.

Table 2. Midpoint environmental impact results for the baseline scenario.

Categories	Abbreviation	Unit	Results
Climate change	GWP	kg CO_2 eq.	7.75×10^2
Fine Particulate Matter Formation	FPMFP	Kg $PM_{2.5}$ eq.	2.84×10^{-1}
Fossil depletion	FDP	kg oil eq.	3.11×10^2
Freshwater Consumption	FWCP	m^3	6.27×10^0
Human toxicity, non-cancer	HTP non	kg 1,4-DB eq.	1.38×10^1
Land use	LUP	Annual crop eq.·y	3.32×10^1
Marine ecotoxicity	METP	kg 1,4-DB eq.	1.45×10^{-1}
Photochemical Ozone Formation	POFP	kg NO_x eq.	8.23×10^{-1}
Terrestrial Acidification	TAP	kg SO_2 eq.	9.02×10^{-1}
Terrestrial ecotoxicity	TETP	kg 1,4-DB eq.	1.79×10^2

FDP (3.11×10^2 kg oil eq.) suggests a need for renewable energy integration, while FWCP (6.27×10^0 m^3) points to the importance of efficient technologies and recycling. HTP (1.38×10^1 kg 1,4-DB eq.) and LUP (3.32×10^1 annual crop eq.) highlight risks mitigated by emission controls and optimized facility design. METP (1.45×10^{-1} kg 1,4-DB eq.) and POFP (8.23×10^{-1} kg NO_x eq.) underscore the necessity for wastewater treatment and improved combustion. The analysis stresses mitigating TAP (9.02×10^{-1} kg SO_2 eq.) and TETP (1.79×10^2 kg 1,4-DB eq.) through neutralization and safe residue disposal. Sustainability can be enhanced through energy efficiency, renewable energy adoption, and waste management.

Normalized midpoint assessment was carried out using the equivalence rate and factors of each selected midpoint category from GaBi software. Each impact category has a different equivalence rate and factor, which were taken from the literature (GaBi

database). Midpoint results were multiplied with relevant factors to obtain normalized midpoint results. Figure 3 shows that the highest contribution/results came from FDP (0.316), indicating high nonrenewable energy use. GWP followed with 0.097 for climate change impacts. POFP came out third with 0.045 due to NO_x and SO_x emissions. TAP, HTP, FWCP, LUP, METP, FPMFP, and TETP were evaluated as 0.022, 0.025, 0.024, 0.005, 0.003, 0.011, and 0.012, respectively. METP was evaluated as having the lowest impact. FPMFP and TETP have comparatively small impacts but are important for ecological sustainability. Those findings identify potential areas for process improvement like fossil fuel dependency reduction, emission optimization, and resource efficiency enhancement.

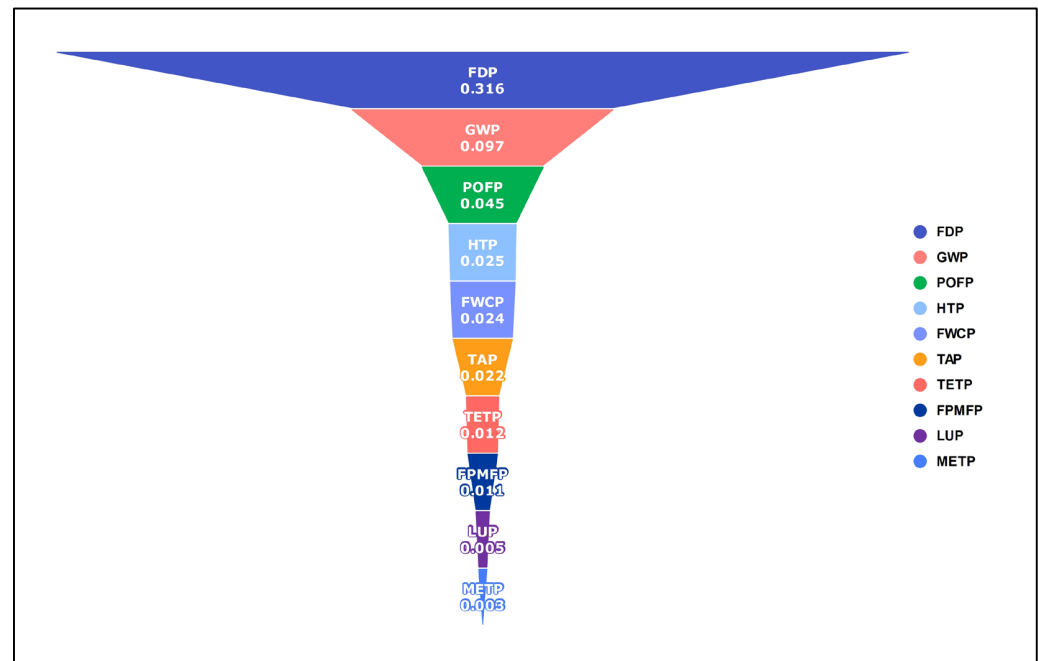


Figure 3. Normalized midpoint impacts for the baseline scenario.

3.2. Endpoint Assessment

Endpoint results from the baseline scenario for plastic waste gasification convey a higher-level view of the environmental impacts of the process, converting midpoint results to impacts on human health, ecosystems, and resources. In terms of resources, the total impact was evaluated as USD 47.0, driven mainly by FDP at USD 46.8 and other categories (metal depletion) at USD 0.223. The dependence of the gasification system on nonrenewable fossil resources is shown to be important for a sustainable development strategy aiming at reducing fossil inputs, as shown in Table 3.

The total impact on human health was 9.13×10^{-4} Disability-Adjusted Life Years (DALY), with the greatest share, 7.21×10^{-4} DALY, from GWP. Other human health categories, FPMFP, HTP, and others, were calculated as 1.79×10^{-4} , 3.16×10^{-6} , and 9.45×10^{-6} DALY, respectively. The HTP value showed very minimal endpoint results due to less exposure to harmful pollutants in this category. These results highlight the need for emissions control technologies to limit air quality impacts and the health burden of the gasification process.

Table 3. Endpoint environmental impacts of syngas production for the baseline scenario.

Categories	Units	Results
Resources Depletion (Total Impact)	USD	4.70×10^1
Fossil depletion	USD	4.68×10^1
Other Categories	USD	2.23×10^{-1}
Human Health (Total Impact)	DALY	9.13×10^{-4}
Climate change	DALY	7.21×10^{-4}
Fine Particulate Matter Formation	DALY	1.79×10^{-4}
Human toxicity, non-cancer	DALY	3.16×10^{-6}
Other Categories	DALY	9.45×10^{-6}
Ecosystem (Total Impact)	species.yr	2.79×10^{-6}
Freshwater Consumption	species.yr	3.17×10^{-8}
Land use	species.yr	2.95×10^{-7}
Marine ecotoxicity	species.yr	1.53×10^{-11}
Terrestrial Acidification	species.yr	1.91×10^{-7}
Terrestrial ecotoxicity	species.yr	2.05×10^{-9}
Photochemical Ozone Formation	species.yr	1.03×10^{-7}
Other Categories	species.yr	2.17×10^{-6}

The total impact on the ecosystem was estimated at 2.79×10^{-6} species.yr, the LUP value at 2.95×10^{-7} species.yr, and the TAP value at 1.91×10^{-7} species. yr. The FWCP value was accounted at 3.17×10^{-8} species.yr. TETP and METP values were calculated as 2.05×10^{-9} species.yr and 1.53×10^{-11} , respectively. METP and TETP values were relatively negligible, reflecting lower ecosystem exposure to harmful substances. These findings suggest that, even though overall ecosystem impacts are relatively low, land use/acidification remains an issue of concern. Change to renewable energy and improved waste management could reduce these impacts and align gasification with other sustainability targets. In covering all these endpoint categories, this study identifies potential for improving the environmental sustainability of plastic waste gasification.

3.3. Hotspot Analysis

The hotspot analysis (Figure 4) shows that electricity consumption is the largest contributor to most impact categories, with 53%, 65%, and 53% of impacts for GWP, FPMFP, and FDP, respectively. Electricity use also has significant impacts on FWCP at 59%, LUP at 72%, and METP at 66%, respectively. These large percentages indicate an energy-intensive gasification process during feedstock preprocessing and reactor operations. Transitioning to renewable sources like solar could reduce these impacts while achieving sustainability goals due to decarbonizing energy inputs. The catalysts, which included air, oxygen, nitrogen, and water required for gasification processes, make them the second largest contributor in most categories, 41% in FWCP and 28% in LUP and HTP. They also contribute 26% to 28% to TAP and TETP, respectively. Steam contributes less to overall impacts but is relevant in GWP (26%) and FDP (26%) because it helps maintain gasification conditions. Together, these results highlight the need for process optimizations like energy efficiency improvement, recycling catalysts, and reduction in steam usage to deal with environmental hotspots and ensure syngas production sustainability.

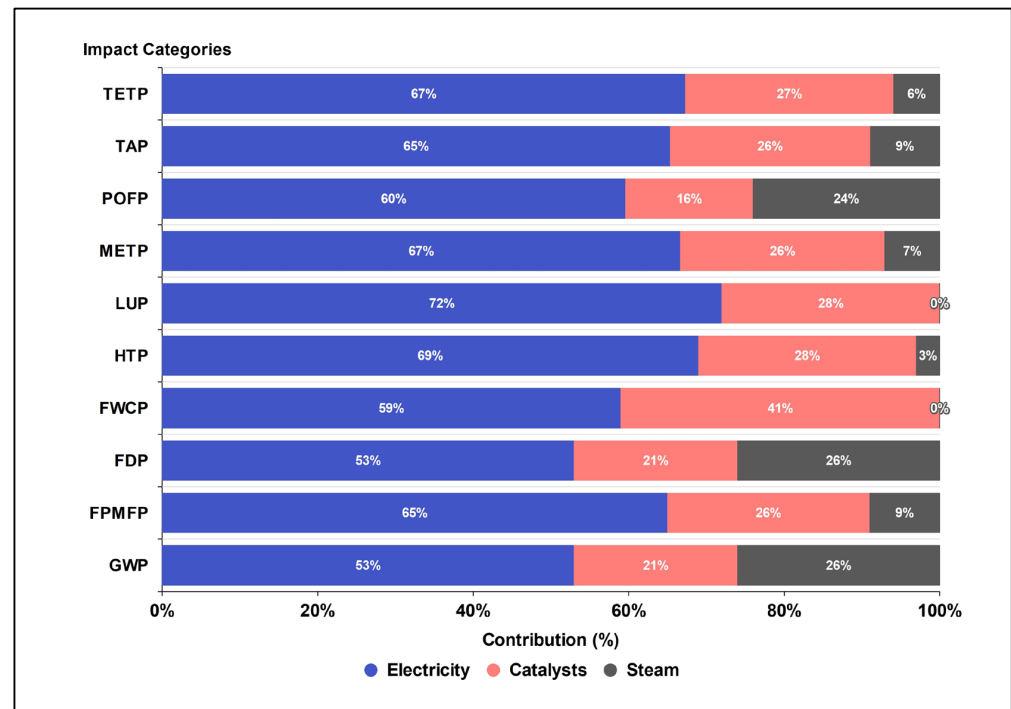


Figure 4. Hotspot analysis for baseline scenario contributors.

3.4. Uncertainty Analysis Results

3.4.1. Sensitivity Analysis

Sensitivity analysis results revealed the variation in environmental impacts due to an increase and decrease of 10% in critical inputs like electricity and operating materials like air, oxygen, nitrogen, steam, water, and sodium hydroxide (Table 4). Electricity is the biggest contributor, with a $\pm 10\%$ change bringing GWP $\pm 5.31\%$, FPMFP $\pm 6.51\%$, FDP $\pm 5.34\%$, FWCP $\pm 7.18\%$, HTP $\pm 6.14\%$, LUP $\pm 7.15\%$, POFP $\pm 5.96\%$, and TAP $\pm 6.56\%$. There are two impact categories, METP and TETP, that did not show any fluctuation from $\pm 10\%$ variation in main contributors. FWCP showed the highest fluctuation compared to the only electricity variations. From sensitivity analysis, it can be determined that electricity is the main contributor to the gasification process. Such increases reflect the energy-intensive processes involved in syngas production and the need to move to alternative electricity sources to reduce environmental impacts.

Table 4. Sensitivity analysis results for baseline scenario impacts.

	Variation	GWP	FPMFP	FDP	FWCP	HTP	LUP	METP	POFP	TAP	TETP
Air	$\pm 10\%$	$\pm 0.17\%$	$\pm 0.21\%$	$\pm 0.17\%$	$\pm 0.23\%$	$\pm 0.19\%$	$\pm 0.23\%$	0.00%	$\pm 0.19\%$	$\pm 0.21\%$	0.00%
Nitrogen	$\pm 10\%$	$\pm 0.32\%$	$\pm 0.39\%$	$\pm 0.32\%$	$\pm 0.43\%$	$\pm 0.37\%$	$\pm 0.43\%$	0.00%	$\pm 0.36\%$	$\pm 0.39\%$	0.00%
Oxygen	$\pm 10\%$	$\pm 1.55\%$	$\pm 1.90\%$	$\pm 1.56\%$	$\pm 2.10\%$	$\pm 1.80\%$	$\pm 2.09\%$	0.00%	$\pm 1.74\%$	$\pm 1.91\%$	0.00%
Sodium Hydroxide	$\pm 10\%$	$\pm 0.07\%$	$\pm 0.08\%$	$\pm 0.05\%$	$\pm 0.03\%$	$\pm 0.07\%$	$\pm 0.09\%$	0.00%	$\pm 0.11\%$	$\pm 0.07\%$	0.00%
Steam	$\pm 10\%$	$\pm 2.59\%$	$\pm 0.91\%$	$\pm 2.57\%$	$\pm 0.03\%$	$\pm 1.39\%$	$\pm 0.02\%$	0.00%	$\pm 1.64\%$	$\pm 0.86\%$	0.00%
Water	$\pm 10\%$	0.00%	0.00%	0.00%	0.00%	$\pm 0.04\%$	0.00%	0.00%	0.00%	0.00%	0.00%
Electricity	$\pm 10\%$	$\pm 5.31\%$	$\pm 6.51\%$	$\pm 5.34\%$	$\pm 7.18\%$	$\pm 6.14\%$	$\pm 7.15\%$	0.00%	$\pm 5.96\%$	$\pm 6.56\%$	0.00%

Steam also showed significant impacts, with a $\pm 10\%$ change resulting in GWP $\pm 2.59\%$, HTP $\pm 1.39\%$, and FDP $\pm 2.57\%$, demonstrating its importance for sustaining reactor con-

ditions. Operating materials, usually oxygen and nitrogen, cause incremental changes. Oxygen usage showed a $\pm 1.55\%$ change in GWP, $\pm 2.10\%$ change in FWCP, and $\pm 2.09\%$ change in TAP. Air, nitrogen, and sodium hydroxide showed minimal changes in selected categories with $\pm 10\%$ input variations. Water showed no impacts with $\pm 10\%$ variation, except in one category, HTP $\pm 0.04\%$. Uncertainty analysis demonstrated that process optimizations such as energy consumption reduction and catalyst efficiency improvement are needed to reduce environmental uncertainties and improve system performance.

3.4.2. Monte Carlo Analysis

The Monte Carlo uncertainty analysis assessed the variability and confidence in midpoint results across selected impact categories, as shown in Table 5. The default values align with the average values, reflecting consistency in the modeling. Standard deviations (SD) range from 4.89% for GWP to 5.78% for HTP, indicating moderate variability across categories.

Table 5. Monte Carlo uncertainty analysis for midpoint impacts.

Categories	Unit	Default	Average	SD	10%	90%
GWP	kg CO ₂ eq.	775	775	4.89%	748	824
FPMFP	kg PM _{2.5} eq.	0.284	0.284	5.46%	0.272	0.306
FDP	kg oil eq.	311	311	5.36%	299	336
FWCP	m ³	6.27	6.27	5.23%	5.97	6.75
HTP	kg 1,4-DB eq.	13.8	13.8	5.78%	13.2	14.9
LUP	Annual crop eq.·y	33.2	33.2	5.39%	31.7	35.5
METP	kg 1,4-DB eq.	0.145	0.145	5.56%	0.139	0.157
POFP	kg NO _x eq.	0.797	0.797	5.10%	0.765	0.856
TAP	kg SO ₂ eq.	0.902	0.902	5.49%	0.863	0.97
TETP	kg 1,4-DB eq.	179	179	5.65%	171	193

The 10th and 90th percentile ranges illustrate the distribution of results, highlighting potential lower and upper bounds of uncertainty. For example, GWP spans from 748 to 824 kg CO₂ eq., FDP ranges from 299 to 336 kg oil eq., and FWCP varies between 5.97 and 6.75 m³. These findings underline the robustness of the model while accommodating data uncertainties. The relatively low variation suggests confidence in the overall outcomes, though categories like HTP and METP may benefit from further data refinement for greater accuracy.

3.5. Scenario Analysis

3.5.1. Midpoint Impacts Comparison

In scenario analysis, two scenarios were compared to evaluate the change in environmental impacts, as shown in Figure 5. One is a baseline scenario in which electricity is taken from a grid mix, while, in the second scenario, electricity is taken from solar. The only difference between these two scenarios was electricity consumption for the gasification process. Scenarios were compared based on electricity sources. The baseline scenario with nonrenewable grid electricity and the renewable scenario with solar electricity showed significant differences in environmental impacts.

The solar scenario showed significant reductions in six selected key categories, including GWP, FPMFP, FDP, LUP, and TAP. The solar scenario, which included four other categories, HTP, METP, POFP, and TETP, showed higher impacts than the baseline scenario. GWP declined 44%, from 775 kg CO₂-eq. in the baseline scenario to 435 kg CO₂-eq. in the solar scenario. This decrease highlights the carbon savings when fossil-based electricity is replaced by solar power. Similar reductions are seen for FPMFP and FDP of 41% and 47%,

respectively, demonstrating the environmental benefit of switching to a renewable energy resource. FWCP is reduced by 50%, which highlights the water-efficiency advantage of solar energy integration.

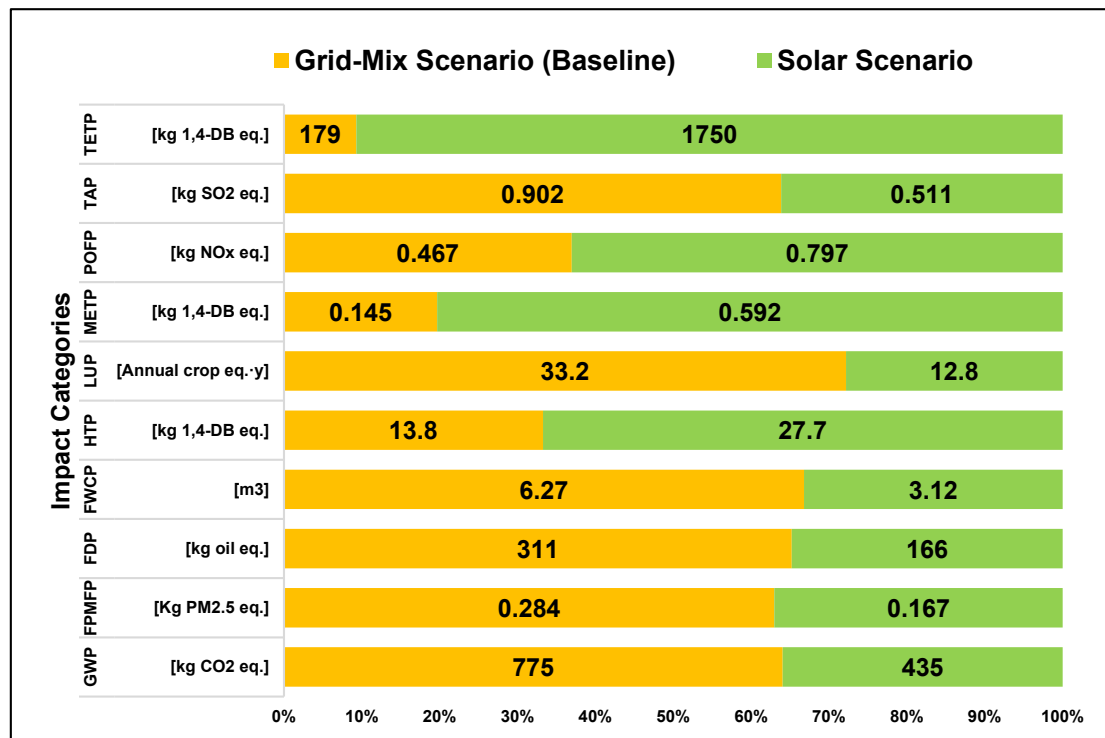


Figure 5. Scenario comparison of midpoint environmental impacts.

However, some categories have higher impacts in the solar scenario. HTP increased by 50% and METP and MDP increased by 76% and 56%, respectively. This is due to the material and resource demands of solar panel manufacturing and infrastructure development, which drive these impact categories. However, LUP decreased by 61%, reflecting lower spatial demands of solar-based electricity generation compared to fossil fuel extraction for grid mix electricity. TAP also decreased by 43%, indicating less acidifying compound emissions in the solar scenario.

3.5.2. Normalized Midpoint Comparison

Normalized midpoint values demonstrated significant differences in environmental impacts across various categories, as shown in Figure 6. The solar scenario revealed lower values for critical categories such as GWP, FPMFP, and FDP. GWP, for instance, decreased from 0.097 in the baseline to 0.054 in the solar scenario, representing a notable reduction in GHG emissions. Similarly, FDP is reduced from 0.316 to 0.169, underscoring the decreased reliance on fossil fuels when transitioning to solar energy. However, tradeoffs are observed in categories such as METP and HTP, where impacts increase from 0.003 to 0.014 and 0.014 to 0.025, respectively, in the solar scenario. FDP resulted in the highest contributing category for both scenarios. The second highest category in the baseline scenario was GWP and, in the solar scenario, it was TETP. Overall, while the solar scenario demonstrates substantial environmental benefits in terms of carbon emissions and resource depletion, the observed tradeoffs highlight the importance of considering the life cycle impacts of renewable technologies.

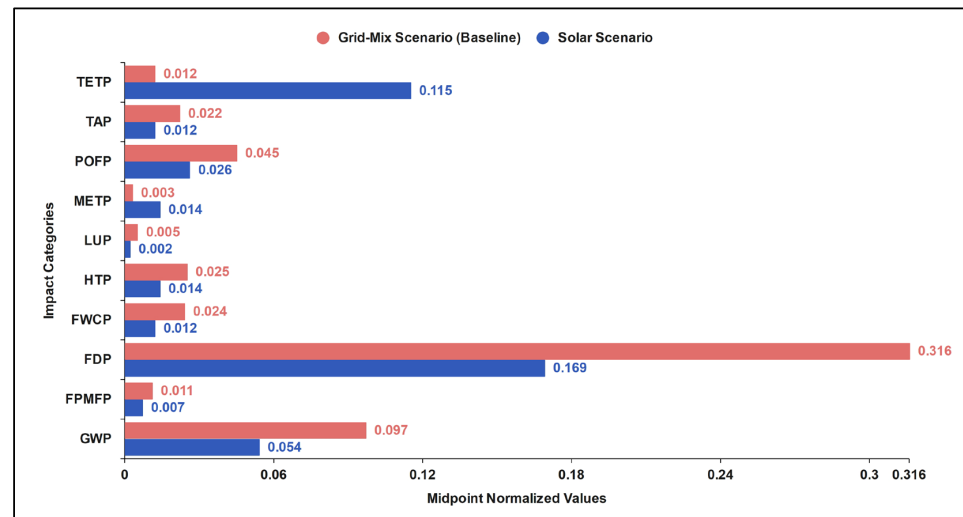


Figure 6. Normalized midpoint values for baseline and alternative scenario.

3.6. Contribution Analysis

The contribution analysis compared the relative environmental impacts of electricity and operating materials in both the grid mix scenario (baseline) and the solar scenario, as shown in Figure 7. In the grid mix scenario, electricity is the dominant contributor across most impact categories, particularly for GWP, FDP, and FPMFP, with contributions of 53%, 65%, and 35%, respectively. Operating materials, which include air, oxygen, nitrogen, steam, water, sodium hydroxide, and other inputs, accounted for a significant share in categories like METP (66%) and TETP (67%). In contrast, the solar scenario showed a significant reduction in electricity contribution, particularly for GWP 16% and FDP 9%, reflecting the environmental benefits of renewable energy integration. However, the reliance on operating materials becomes more apparent, contributing 84% to GWP and 91% to FDP. Similarly, operating materials dominated HTP (84%) and METP (92%), driven by resource-intensive solar panel production. These results underscore the tradeoffs between transitioning to renewable energy and increased material demands, highlighting the need for life cycle optimizations in solar infrastructure to maximize sustainability.

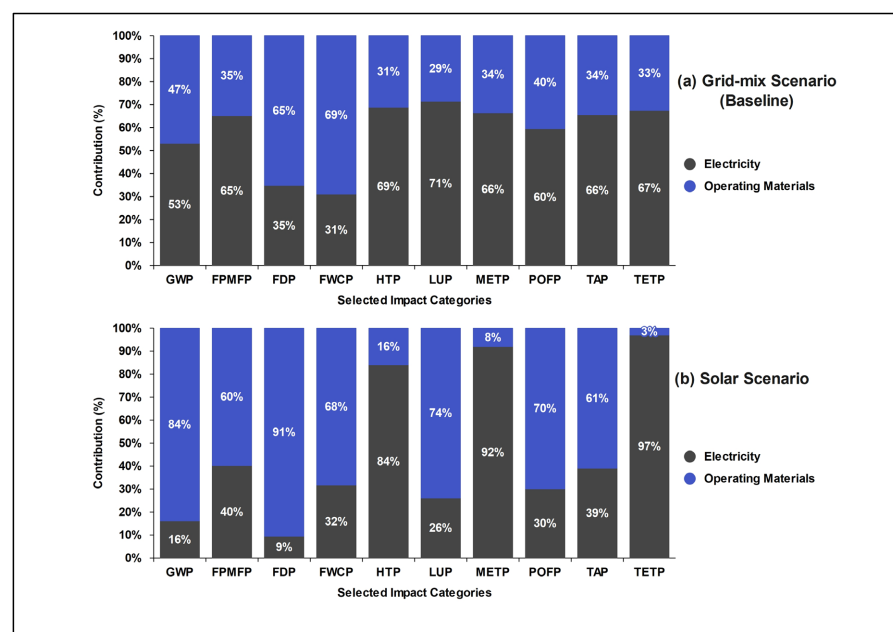


Figure 7. Contribution analysis of inputs to impacts. (a) Grid mix scenario; (b) solar scenario.

3.7. Life Cycle Costing of Gasification Plant

LCC was calculated using Equation (6), as mentioned in the methodology section. For IC, case study is mandatory to calculate different calculations for the gasification plant. For that reason, it is more reasonable to target Pakistan as a country and then further Lahore. Pakistan's plastic waste crisis, which generates an average of 3.3 million tons of waste per year, is an enormous challenge for sustainable waste management [48]. Lahore is among the biggest contributors, with an average of 700 tons of plastic waste a day [49]. Lahore Waste Management Company (LWMC) signed an agreement with the DG Khan cement industry in 2011 to supply MSW for 15 years [50]. As a result, a material recovery facility (MRF) was established at Lakhodair, an area located in the north of Lahore. The facility has a waste processing capacity of 1000 tons per day. It means Lahore is already halfway to reducing plastic waste so that it would be a more cost-saving strategy for Lahore's plastic waste.

Besides recycling, the remaining nonrecyclable mixed plastic waste can be recovered as energy by gasification. As a case study, this analysis examines the economic viability of a 50 tons/day (TPD) gasification plant at Lahore for the conversion of mixed plastic waste into electricity by a combined heat/power system. This plant would process 50 tons of plastic waste for eight hours daily, handling 1500 tons of plastic waste monthly. Gasification of this waste yields 1998 Nm³ of syngas per ton plastic waste (according to Table 1 inventory data), which is converted into electricity via a combined heat and power (CHP) system that produces 1850 kWh/ton of plastic waste. It will produce 33,762,500 kWh of electricity annually. At the USD 0.142/kWh market selling price, electricity sales are expected to generate USD 4,794,275 in revenue per year (Table 6).

Table 6. Internal cost data for the 50 TPD gasification plant.

Component	Economic Assumptions
1. Depreciation	Life period 20 years; Salvage value 5% [46]
2. Equipment cost	USD 1.5 million (Quotes by equipment supplier after customization)
3. Construction cost	USD 36000 (Local Market rate)
4. Raw materials	Oxygen USD 0.04/Nm ³ , Nitrogen USD 0.1/Nm ³ , Caustic soda (sodium hydroxide USD 0.5/kg, Steam USD 0.01/kg, Air USD 0.02/Nm ³ (Local Market rate)
5. Utilities cost	Electricity USD 0.126/kWh (LESCO rate), water USD 0.0036/kg
6. Management Cost	USD 50,000 per Annum (LWMC staff)
7. Labor cost	USD 21,660 per Annum (10 workers and 1 manager) Local wages rate
8. Maintenance cost	USD 50,000 per annum (LWMC staff)

The plant's ICI was calculated at USD 1.536 million, including construction and equipment procurement. Depreciation over 20 years of this investment at 5% salvage value by the straight-line method causes USD 72,960 per year in depreciation cost. Annual operating expenditures (OPEX) were calculated at USD 4,048,538, which constituted USD 1,513,323 for raw materials such as oxygen, nitrogen, steam, and sodium hydroxide, USD 2,342,304 for utilities such as electricity and water, USD 50,000 for management, USD 21,661 for labor, and USD 50,000 for maintenance. Other ECs related to emissions are calculated using Table 7. A total EC of USD 26.43 per ton of plastic waste processed is evaluated. The annual EC of a 50 TPD plant is evaluated as USD 475,740. Such ECs cover emissions like CO₂, CO, NO_x, PM, NMVOC, and SO₂, which show the environmental tradeoffs of gasification. Despite these costs, avoided emissions and reduced landfill dependency provide significant environmental benefits in line with sustainability targets.

Table 7. External cost analysis for baseline scenario emissions.

Coefficient		Baseline Scenario		Contribution
Elements ^a	\$/kg ^b	Emissions (Kg) ^c	EC (USD)	%
CO ₂	0.028	7.17×10^2	2.01×10^1	76
CH ₄	0.21	1.35×10^0	2.84×10^{-1}	1.1
NO _x	4.47	7.75×10^{-1}	3.46×10^0	13.1
CO	0.59	5.02×10^{-1}	2.96×10^{-1}	1.1
NMVOC	1.24	1.10×10^{-1}	1.36×10^{-1}	0.5
PM	10.13	1.44×10^{-2}	1.46×10^{-1}	0.6
SO ₂	3.41	5.95×10^{-1}	2.03×10^0	1.1
Total		26.43 USD/ton		100%

^{a,b} Elements and coefficient values taken from the literature [51]. ^c Emissions values taken from GaBi.

Key metrics are used to evaluate the financial viability of the plant. With USD 4,794,275 annual revenue and USD 4,048,538 annual OPEX, the net profit is USD 745,737 per year. The payback period (PB) is approximately 2.06 years, which indicates a rapid recovery in investment. Net present value (NPV) for a 20-year project life at a 10% discount is calculated using Equation (10) as follows:

$$NPV = \sum_{t=1}^{20} \frac{\text{Net Cash Flow}}{(1+r)^t} - \text{Initial Investment} \quad (10)$$

Substituting USD 745,737 cash flow and a 10% discount rate calculated an NPV of approximately USD 5.32 million. This positive NPV indicates that the project is economically viable in the long term. In addition to the profitability of the project, the internal rate of return (IRR) at which the NPV becomes zero is 38.2%, far above the 10% discount rate. Syngas conversion to electricity is carried out by a CHP system integrated into the plant for economic and efficiency reasons. The system captures electricity and thermal energy to maximize energy recovery from syngas and minimize energy losses. This technology contributes to plant sustainability by enhancing energy efficiency and emission reductions compared with conventional energy systems. CHP also fits in with global trends of sustainable energy production based on resource optimization and environmental stewardship.

Economic analysis showed that a 50 TPD gasification plant in Lahore is financially feasible, with a short payback time, high IRR, and high NPV. A CHP system is also integrated into the plant for improved economic and environmental performance. ECs highlight emissions management needs, but the plant's clean energy potential and landfill dependency make the investment case strong. If supported by subsidies and tax incentives, the scalability of such projects could change waste management in Pakistan towards environmental sustainability and energy security.

4. Discussion

4.1. Comparative Analysis with Existing Studies

This study adds substantial value to the increasing research on WTE technologies by comprehensively evaluating gasification's environmental and economic benefits. Compared to traditional incineration, gasification offers superior energy recovery efficiency and significant reductions in GHG emissions, making it a cleaner alternative [52,53]. Additionally, unlike incineration, gasification reduces harmful byproducts like dioxins and lessens greenhouse gas emissions through controlled partial oxidation. Compared to pyrolysis, which often demands extensive feedstock sorting, gasification is more tolerant of mixed and contaminated waste streams, enhancing its applicability to real-world waste scenarios [12].

Gasification, in comparison to pyrolysis, shows greater energy recovery efficiency because it can handle mixed plastic waste without requiring extensive preprocessing. Additionally, this study's techno-economic analysis (TEA) underscores the cost-effectiveness of syngas production from plastic waste, especially when feedstock costs are minimized through supportive policies such as subsidized collection systems [54]. By combining LCA with TEA, this study fills a critical gap in the literature, providing actionable insights into gasification's role in achieving sustainable plastic waste management within a circular economy framework [12,46]. Table 8 compares different LCA studies with this study.

Table 8. Comparative study of gasification and similar LCA studies.

Study	Functional Unit	Technology	Feedstock Type	Methodology	Key Findings
This Study	1 ton of plastic waste	Steam and Oxygen gasification	Mixed plastic waste	ReCiPe 2016	44% reduction in GWP using solar energy, high sensitivity to electricity inputs.
[52]	1 MJ of naphtha	Pyrolysis	Nonrecyclable plastic waste	Mass Allocation	Reduce plastic waste but environmental emissions
[53]	1 kg of methanol and hydrogen	Steam gasification	Mixed plastic waste	TRACI 2.1 (2008)	Feedstock cost and syngas yield is critical to economic and environmental performance
[54]	1 kg of syngas	Co-gasification with biomass	Mixed plastics and biomass	Environmental Footprint (EF) 3.0	Co-gasification lowers emissions and improves hydrogen-to-carbon ratios.
[55]	2.2 tonnes of plastic waste	Landfilling vs. Energy recovery	Mixed Plastic waste	ReCiPe 2016	Landfilling impacts a lot more than energy recovery emissions

4.2. Recent Innovations in Gasification Technology

Recent advancements in gasification technology have significantly improved efficiency and operational flexibility, enhancing the feasibility of large-scale adoption. Catalytic gasification, which uses catalysts like nickel and iron composites, reduces tar formation and boosts hydrogen production, delivering higher-quality syngas [56,57]. Plasma-assisted gasification, another promising innovation, employs high-temperature plasma arcs to convert mixed and contaminated plastic waste into syngas efficiently, with minimal emissions [58]. Additionally, co-gasification with biomass has emerged as a successful strategy to optimize the hydrogen-to-carbon ratio, capitalizing on the complementary properties of plastic and biomass feedstocks [54]. These innovations address challenges such as feedstock variability and impurity removal while improving overall process efficiency and environmental performance.

4.3. Policy Insights

Gasification technologies need robust policy frameworks to encourage their adoption and scalability when integrated into modern waste management systems. Carbon credits may be a practical way of rewarding GHG emission reductions, since syngas production by gasification offsets the negative environmental impacts of landfilling and incineration of conventional waste management practices [56]. Decarbonizing electricity inputs and lowering operational costs, tax benefits, and subsidies for renewable energy use, such as

solar and wind in gasification plants, may enhance environmental outcomes further [52]. These policy approaches are in line with global commitments such as the Paris Agreement and Sustainable Development Goals, in particular, SDG 7—affordable and cleaner energy and SDG 13—climate action, which call for cleaner energy systems and a reduction in dependency on fossil fuels [34,53].

In addition to financial incentives, structured waste management policies are needed to ensure a consistent feedstock supply. Segregation and preprocessing policies might reduce operation inefficiencies, improve resource recovery, and promote adherence to circular economy principles. Public Private Partnerships may be an important instrument for enabling funding and operation support whilst sharing the risks of large-scale gasification facilities. Emission control and standard practices for gasification processes also need to be regulated by policymakers for environmental compliance. Developing regional and national strategies to integrate gasification technologies into waste-to-energy frameworks will maximize resource recovery, minimize environmental impacts, and position gasification as a key technology in modern, sustainable waste management systems.

4.4. Sustainability Implications

Implications for the sustainability of gasification technologies for plastic waste are diverse in environmental, economic, and social dimensions. The gasification reduces landfill dependency and methane and dioxin emissions associated with traditional waste management [12]. Changing mixed and contaminated plastics to high-value syngas supports resource optimization and circular economy principles [9]. Gasification generates revenue economically through syngas utilization for electricity production, synthetic fuels, and chemical manufacturing. It also reduces societal costs of waste mismanagement, such as the health costs of pollution.

Socially, gasification facilities may create jobs in waste collection, plant operation, and syngas distribution. In addition, gasification plants are scalable to meet urban and rural waste management challenges worldwide. Sustainable development also requires constant innovation. Recently developed technologies like catalytic gasification and plasma-assisted systems increase syngas yield and energy efficiency, making the technology competitive with conventional energy sources [57]. Gasification needs policy alignment and international co-operation to become a mainstream solution to manage plastic waste sustainably.

Plastic waste valorization has a strong link with SDG 7, SDG 12, and SDG 13. Gasification transforms plastic waste into syngas for electricity generation, specifically supporting SDG 7, which mainly focuses on affordable and clean energy. D'Adamo et al. [59] highlighted the importance of waste-to-energy (WTE) technologies that generate sustainable energy. Plastic waste gasification also contributes to SDG 12 (responsible consumption and production), which emphasizes circular economy approaches in which plastic waste is turned into energy carriers rather than being landfilled. Furthermore, for SDG 13 climate action, greenhouse gas emissions from conventional waste disposal are reduced. This study expands on these perspectives by evaluating gasification as a plastic waste valorization strategy to fill the knowledge gap in LCA-based environmental and economic assessments of this technology.

4.5. Practical Implications

Plastic waste gasification technologies, in practice, open new perspectives for waste management and energy production. Gasification addresses mixed and contaminated plastic streams and is a scalable solution to existing waste management infrastructure in urban centers like Lahore. Mixed plastic waste presents challenges due to variations in reactivity and thermal decomposition properties of different polymers. In this study, the

gasification process was designed to handle these variations through high temperature (800–1200 °C) to ensure consistent decomposition of various polymer types into syngas. This approach also reduces the demand for feedstock segregation and minimizes waste handling costs. Establishing new gasification plants with different capacities allows for the handling of different waste volumes in diverse geographic areas, including rural areas, without traditional waste management systems. In addition, gasification facilities, in combination with MRFs, improve preprocessing efficiency and feedstock availability. Technology and operation strategies must align to ensure future gasification success. For example, incorporating renewable energy sources such as solar or wind into gasification plant electricity demands may further decarbonize operations while reducing dependency on fossil fuels. With CHP systems, the syngas energy potential is maximized by the dual generation of electricity and thermal energy. This improves process efficiency and makes gasification competitive with conventional energy systems.

Gasification requires targeted investments in R&D to refine process parameters and reduce costs. Collaboration between academia, industry, and government can result in catalyst development, feedstock preprocessing, and emission control systems innovations. International collaborations can exchange experiences and standardize practices, allowing gasification to become a key technology in the global transition to sustainable energy and waste management systems. Finally, practical implementation requires workforce training and community engagement. Stakeholder education regarding gasification benefits and plastic waste crises will increase public acceptance and participation in waste segregation activities. Prioritizing these practical steps will allow gasification to achieve theoretical feasibility in addition to real-world impact to provide a long-term, resilient, and economically viable framework for waste-to-energy conversion.

4.6. Study Limitations

This study depends on secondary data sources for LCA inventory, which may present variability in results. This study assumed a consistent composition for syngas, which may vary in real-world scenarios. Additionally, the economic evaluation focuses on syngas and electricity production without accounting for potential byproduct valorization, such as char or slag reuse. Future studies should examine dynamic LCA approaches and incorporate regional variability in energy inputs to improve the robustness of the findings.

5. Conclusions

This work demonstrates that plastic waste gasification can be a cutting-edge and sustainable solution to the twin problems of waste management and energy generation. This study aligns with multiple SDGs like SDG 7, SDG 12, and SDG 13. The analysis using Lahore as a case study provides insights on gasification technology integration into urban waste management frameworks. A 50-ton-per-day (TPD) gasification plant was evaluated considering the city's annual plastic waste generation of about 700 tons. The LCA shows that gasification leads to significant GHG reductions (SDG 13), reduced landfill dependency, and compliance with circular economy principles (SDG 12). These findings place gasification among cleaner alternatives to landfilling/incineration (SDG 7). From an economic point of view, the analysis identifies the plant's profitability, which is supported by a short PB and high NPV/IRR. CHP systems for electricity recovery from syngas are further enhanced to make the plant economically and environmentally sustainable. ECs such as emissions remain to be considered, but the overall feasibility of gasification demonstrates its potential to change waste management in urban centers. It confirms gasification as an important technology for plastic pollution reduction, renewable energy transition, and global sustainability goals.

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